

Influence of Strength on Magnitude and Mechanisms of Adaptation to Power Training

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ABSTRACT

CORMIE, P., M. R. MCGUIGAN, and R. U. NEWTON. Influence of Strength on Magnitude and Mechanisms of Adaptation to Power Training. *Med. Sci. Sports Exerc.*, Vol. 42, No. 8, pp. 1566–1581, 2010. **Purpose:** To determine whether the magnitude of performance improvements and the mechanisms driving adaptation to ballistic power training differ between strong and weak individuals. **Methods:** Twenty-four men were divided into three groups on the basis of their strength level: stronger ($n = 8$, one-repetition maximum-to-body mass ratio (1RM/BM) = 1.97 ± 0.08), weaker ($n = 8$, 1RM/BM = 1.32 ± 0.14), or control ($n = 8$, 1RM/BM = 1.37 ± 0.13). The stronger and weaker groups trained three times per week for 10 wk. During these sessions, subjects performed maximal-effort jump squats with 0%–30% 1RM. The impact of training on athletic performance was assessed using a 2-d testing battery that involved evaluation of jump and sprint performance as well as measures of the force–velocity relationship, jumping mechanics, muscle architecture, and neural drive. **Results:** Both experimental groups showed significant ($P \leq 0.05$) improvements in jump (stronger: peak power = $10.0 \pm 5.2 \text{ W} \cdot \text{kg}^{-1}$, jump height = $0.07 \pm 0.04 \text{ m}$; weaker: peak power = $9.1 \pm 2.3 \text{ W} \cdot \text{kg}^{-1}$, jump height = $0.06 \pm 0.04 \text{ m}$) and sprint performance after training (stronger: 40-m time = $-2.2\% \pm 2.0\%$; weaker: 40-m time = $-3.6\% \pm 2.3\%$). Effect size analyses revealed a tendency toward practically relevant differences existing between stronger and weaker individuals in the magnitude of improvements in jump performance (effect size: stronger: peak power = 1.55, jump height = 1.46; weaker: peak power = 1.03, jump height = 0.95) and especially after 5 wk of training (effect size: stronger: peak power = 1.60, jump height = 1.59; weaker: peak power = 0.95, jump height = 0.61). The mechanisms driving these improvements included significant ($P \leq 0.05$) changes in the force–velocity relationship, jump mechanics, and neural activation, with no changes to muscle architecture observed. **Conclusions:** The magnitude of improvements after ballistic power training was not significantly influenced by strength level. However, the training had a tendency toward eliciting a more pronounced effect on jump performance in the stronger group. The neuromuscular and biomechanical mechanisms driving performance improvements were very similar for both strong and weak individuals. **Key Words:** BALLISTIC, JUMP, SQUAT, SPRINT, NEUROMUSCULAR ADAPTATIONS

After strength training, the magnitude of improvements in strength and the mechanisms driving those adaptations differ as the strength level of the athlete improves (14,17,33,37). Specifically, initial improvements in strength are much greater and predominately driven by neural adaptations (although early phase muscular adaptations also occur), whereas further increases in strength are progressively harder to achieve and morphological adaptations in the muscle become more important (14,17,33,37). Thus, training programs geared at significantly improving strength in individuals with an existing

high level of strength require a much more sophisticated design (i.e., greater specificity and variation) (17,33). Although the factors contributing to maximal strength and the application of strength training are well understood, much less is known concerning the adaptations to and utilization of ballistic power training. In particular, the influence of strength level on both the magnitude of improvement and the mechanisms driving adaptation after ballistic power training is not known. Such knowledge is vital to the development of training programs that most effectively improve maximal power production and athletic performance in athletes with a wide variety of training backgrounds.

Cross-sectional comparisons have revealed that individuals with higher strength levels have markedly superior power production capabilities than those with a low level of strength (2,7,10,24,34,35). For example, significant differences in power output and/or jump height between individuals with significantly different strength levels have been reported in comparisons of well-trained athletes and relatively untrained controls (10,24,35), athletes competitive in power-type sports (i.e., volleyball) and endurance events (7), rugby league players involved in national versus

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state competitions (2), as well as the strongest and weakest in a pool of resistance-trained men with various training backgrounds (34). These findings are supported by reports of a significant and positive relationship existing between maximal strength and maximal power production (1,30,34). While skeletal muscle mechanics that cause the force–velocity relationship dictate that maximal strength plays a role in muscular power, the development of muscular power is influenced by a multitude of factors in addition to maximal strength (27). For example, McBride et al. (24) observed that despite no differences in the maximal strength of national-level weightlifters and powerlifters (Smith machine squat one-repetition maximum-to-body mass ratio (1RM/BM) = 2.86 ± 0.15 and 2.88 ± 0.14 , respectively), weightlifters generated significantly greater power output during an unloaded countermovement jump (CMJ; weightlifters = $63.0 \pm 2.7 \text{ W} \cdot \text{kg}^{-1}$, powerlifters = $56.9 \pm 2.5 \text{ W} \cdot \text{kg}^{-1}$). Such observations were attributed to the various training protocols commonly used by these athletes (i.e., high force–high velocity training vs high force–low velocity training common to weightlifting and powerlifting, respectively) (24). Similar results were observed in comparisons between well-trained power athletes and recreational bodybuilders with similar strength levels (35). Therefore, although the development of maximal muscular power required for the successful performance of many athletic movements is influenced by a multitude of factors (27), stronger individuals have consistently been shown to display superior power output than individuals with a significantly lower strength level (2,7,10,24,34,35).

This raises the question of what mechanisms contribute to the improved power production capabilities of stronger individuals. Stronger individuals possess neuromuscular characteristics that form the basis for superior maximal power production and ultimately contribute to enhanced athletic performance. Specifically, an individual with a substantially greater level of strength (i.e., strong vs weak person) would have larger whole-muscle cross sectional area (CSA), a result of greater myofibrillar CSA of both Type I and Type II fibers, with more pronounced hypertrophy of Type II fibers evident (18,22). Pennation angle may also be greater and possibly even fascicle length (3,4,21). In addition, effectiveness of neural drive (i.e., recruitment, rate of onset, firing frequency) as well as intermuscular coordination would be far superior in the significantly stronger individual (15,25,33). These neuromuscular characteristics would result in a shift in the force–velocity relationship so that the force of muscle contraction would be greater for any given velocity of shortening (19). As a consequence, the ability to generate maximal muscular power, and therefore perform athletic movements, would be superior in a considerably stronger individual compared with a weaker person (19,23,38).

Despite the advantage of strength for maximal power production and athletic performance, little is known regarding whether neuromuscular characteristics of stronger individuals allow for superior adaptation to ballistic power

training. More importantly, it is not known if the mechanisms driving adaptations to ballistic power training are influenced by strength level. To the authors' knowledge, only one intervention study examining the possible influence of strength on improvements after ballistic power training exists (37). Wilson et al. (37) compared the improvements between stronger ($n = 4$; squat 1RM/BM = 1.99 ± 0.30) and weaker men ($n = 5$; squat 1RM/BM = 1.21 ± 0.18) after 8 wk of drop jump training. Neither group significantly improved either jump and reach height or 20-m sprint time, and no correlation was found between strength level and the magnitude of the training-induced change in jump and reach height ($r = -0.13$, $n = 14$) (37). Therefore, this previous study offers little insight into the influence of strength on the magnitude of improvements in athletic performance after ballistic power training. Furthermore, the investigation did not involve measures to examine any mechanistic factors possibly involved in adaptations to the training. Simulation data suggest that although jump height is most sensitive to increases in strength-to-BM ratio (31), an increase in strength does not translate into increased jump height unless control (i.e., intermuscular coordination—the appropriate magnitude and timing of activation of agonist, synergist, and antagonist muscles during a movement) is tuned to the strengthened muscle properties (6). These data are supported by the concept of a delayed training effect in which it takes considerable time and effort for an increase in strength to transmute to improved performance in multijoint movements because intermuscular coordination needs to be adapted to the stronger motor units/muscles (39). Therefore, it may be speculated that a stronger individual exposed to ballistic power training would improve performance by fine tuning the timing and patterns of neural drive, thus improving intermuscular coordination and movement technique (although muscular adaptations at the cellular level such as alterations to protein expression, calcium release, and uptake as well as relaxation time may also contribute). It is unknown if the combination of these potential changes with the underlying neuromuscular characteristics of the stronger individual would result in a training response dissimilar to individuals with lower levels of strength.

In comparisons of individuals not currently involved in ballistic power training, previous research has demonstrated that stronger individuals are able to generate superior levels of maximal power than significantly weaker individuals (2,7,10,24,34,35). Although the enhanced maximal power production of stronger individuals in the absence of ballistic power training is theorized to be due to the underlying neuromuscular characteristics of stronger individuals, it is unclear if these characteristics improve the ability to *adapt* to ballistic power training. Specifically, it is not known whether ballistic power training elicits performance improvements in stronger individuals of a greater magnitude than that which is achievable with weaker subjects. Furthermore, there is a paucity of research investigating whether strength level influences the mechanisms responsible for performance

improvements after ballistic power training. Therefore, the purpose of this experiment was to determine whether the magnitude of performance improvements and the mechanisms driving adaptation to ballistic power training differ between strong and weak individuals.

METHODS

Experimental design. This study used a randomized, control design and was conducted during a period of 15 wk. Subjects were divided into two strata on the basis of their 1RM/BM: stronger (1RM/BM > 1.55) or weaker (1RM/BM < 1.55). Subjects in the stronger stratum were allocated into the stronger group. Subjects in the weaker stratum were randomized into one of two groups: weaker group or control group. The stronger and weaker groups completed a total of 10 wk of ballistic power training, while the control group maintained their normal level of activity throughout the duration of the study. Training involved three sessions per week in which subjects performed maximal-effort jump squats with 0%–30% 1RM. Subjects completed a 2-d testing battery before initiating training (baseline), after 5 wk of training (midtest—stronger and weaker groups only), and after the completion of 10 wk of training (posttest). Subjects were adequately familiarized to all testing procedures before actual assessment. Testing involved evaluation of jump and sprint performance as well as measures of the force–velocity relationship, jumping mechanics, muscle architecture, and neural drive.

Subjects. Subjects were recruited on the basis of their ability to perform a back squat with proficient technique. A total of 35 men fulfilled all the testing and training requirements of this investigation. Data from 11 of these men were removed on the basis of their 1RM/BM to establish two experimental groups with very distinct differences in maximal strength (i.e., data from subjects with a 1RM/BM between 1.55 and 1.85 were not included). The remaining 24 men were allocated into three groups: stronger group ($n = 8$, 1RM/BM = 1.97 ± 0.08), weaker group ($n = 8$, 1RM/BM = 1.32 ± 0.14), or control group ($n = 8$, 1RM/BM = 1.37 ± 0.13). Subjects' characteristics throughout the du-

ration of the study are outlined in Table 1. The participants were notified about the potential risks involved and gave their written informed consent. This study was approved by the university's human research ethics committee.

Training program. The training program consisted of three sessions per week separated by at least 24 h of rest. Each training session was initiated via a warm-up consisting of two sets of six submaximal jump squats with 0% 1RM (i.e., CMJ with no external load, just the resistance applied by a carbon fiber pole (mass = 0.4 kg) held across the shoulders). During sessions 1 and 3 of each week, subjects performed seven sets of six maximal-effort jump squats separated by a 3-min recovery. Jump squats were performed at the load that maximized power output for each subject, as determined during the baseline testing session. Similar to previous research, a load consistent with the subject's BM (i.e., no external load or 0% 1RM) maximized power output for each of the participants in this study (11,12). The second training session of each week included an additional warm-up set consisting of five submaximal jump squats with 30% 1RM. Subjects then performed five sets of five maximal-effort jump squats with 30% 1RM separated by a 3-min recovery. Subjects were encouraged to perform each jump as rapidly as possible. Intensity was modified for each session so that an audible beep could be heard by subjects during jumps that reached 95% of the maximal power output at that load from their previous training or testing session. Previous literature has shown significant performance improvements after jump squat training with similar programming parameters at both 0% 1RM (11) and 30% 1RM (25,38). Subjects refrained from any additional lower body resistance training, plyometrics, or sprint training throughout the course of the study.

Testing protocol. Subjects rested for 3–5 d between their previous training session and the midtest testing session and for 7–10 d between the final training session and the posttest testing session to allow full recovery. Subjects completed two testing sessions separated by at least 3 d of recovery at the baseline, midtest, and posttest occasions. Testing session 1 was initiated with an examination of the vastus lateralis (VL) muscle architecture through

TABLE 1. Subject characteristics of the stronger, weaker, and control groups throughout the 10 wk of training.

	BM (kg)	Body Fat (%)	1RM/BM	40-m Sprint (s)	Jump Height (m)
Stronger group					
Baseline	79.1 ± 12.8	12.2 ± 3.9†	1.97 ± 0.08†	5.41 ± 0.25†	0.43 ± 0.03‡
Midtest	79.8 ± 12.9	12.3 ± 3.8‡	1.93 ± 0.11‡	5.39 ± 0.20‡	0.50 ± 0.03*
Posttest	79.6 ± 13.0	12.6 ± 3.5‡	1.88 ± 0.11†	5.29 ± 0.20†	0.50 ± 0.03*§
Weaker group					
Baseline	79.9 ± 14.5	17.3 ± 3.8	1.32 ± 0.14	5.90 ± 0.27	0.38 ± 0.04
Midtest	79.2 ± 14.2	17.5 ± 4.0	1.38 ± 0.16	5.79 ± 0.20	0.42 ± 0.07
Posttest	79.1 ± 13.8	17.7 ± 3.6	1.39 ± 0.17	5.69 ± 0.22	0.44 ± 0.06
Control group					
Baseline	77.5 ± 8.1	14.6 ± 3.5	1.37 ± 0.13	5.79 ± 0.19	0.41 ± 0.04
Posttest	78.4 ± 8.8	14.7 ± 3.8	1.35 ± 0.12	5.79 ± 0.12	0.40 ± 0.04

BM, body mass; 1RM:BM, squat one repetition maximum to body mass ratio.

* Significantly ($P \leq 0.05$) different from baseline.

† Significant ($P \leq 0.05$) difference between stronger group and all other groups.

‡ Significant ($P \leq 0.05$) difference between stronger group and weaker group.

§ Significantly ($P \leq 0.05$) different from control group.

ultrasonography. Maximal dynamic strength was then assessed using a back squat 1RM to a depth consistent with a knee angle of at least 90° of flexion assessed using two-dimensional motion analysis. During a 30-min recovery, body composition was assessed using dual-energy x-ray absorptiometry. Maximal isometric strength was then evaluated using an isometric squat test performed at a knee angle of 140° to allow for the determination of maximum force output at zero velocity (12,32). Adequate recovery was permitted (10 min) before examination of jump squat performance across a series of intensities: 0% (i.e., no external load or BM only), 20%, 40%, 60%, and 80% of squat 1RM. Subjects completed the jump squats in a randomized order, which was consistent across the three testing occasions for each individual subject. Kinematic (both linear position transducer (LPT) and two-dimensional motion analysis), kinetic, and EMG data were obtained simultaneously throughout the testing session. After at least 3 d of recovery, subjects completed the second testing session involving a 40-m sprint test.

Data acquisition and analysis procedures. The back squat 1RM involved subjects completing a series of warm-up sets (four to six repetitions at 30% estimated 1RM, three to four repetitions at 50% estimated 1RM, two to three repetitions at 70% estimated 1RM, and one to two repetitions at 90% estimated 1RM), each separated by 3 min of recovery. A series of maximal lift attempts was then performed until a 1RM was obtained. No more than five attempts were permitted with each attempt separated by 5 min of recovery. This protocol has been frequently used throughout the previous literature for the assessment of maximal dynamic strength (11,12,30). Only trials in which subjects reached a relative knee angle (i.e., angle between the midline of the lower leg and the midline of the thigh) <90° of flexion were considered successful. This depth was visually monitored during testing and was confirmed by two-dimensional motion analysis (stronger group: baseline = 85.7° ± 4.2°, posttest = 82.5° ± 6.8°; weaker group: baseline = 83.4° ± 2.4°, posttest = 80.4° ± 4.3°; control group: baseline = 83.1° ± 6.2°, posttest = 82.6° ± 5.4°; rater reliability: $r = 0.95$). Although no significant differences in the depth of the squat 1RM existed among the testing occasions, there is the potential that the slightly deeper depth obtained during posttesting could have influenced the 1RM values obtained.

The isometric squat test was performed with subjects standing on a force plate (9290AD; Kistler Instruments, Winterthur, Switzerland) in a back squat position pushing against an immovable rigid bar. The bar was positioned so that subjects had a knee angle of 140° of flexion to allow for the determination of maximal force output at zero velocity. Previous research has shown this knee angle to correspond with the highest isometric force output for the squat compared with a range of other knee angles (32). Subjects were instructed to perform a rapid, maximal effort to reach maximal force output as soon as possible and maintain that

force for 3 s. The analog signal from the force plate was collected for every trial at 1000 Hz using a data acquisition system including an analog-to-digital card (cDAQ-9172; National Instruments, North Ryde, NSW, Australia). Custom programs designed using LabVIEW software (Version 8.2; National Instruments) were used for recording and analyzing the data. The signal was filtered using a fourth-order, low-pass Butterworth filter with a cutoff frequency of 50 Hz. From laboratory calibrations, the voltage output was converted into vertical ground reaction force. Peak force relative to BM was assessed as the maximal force output during the 3-s period divided by the individual's BM. The test-retest reliability for peak force relative to BM was $r = 0.98$.

Performance of a jump squat involved subjects completing a maximal-effort CMJ while holding a rigid bar across their shoulders. Subjects held a 0.4-kg carbon fiber pole for the 0% 1RM jump squat, whereas for all other intensities, subjects held a 20-kg barbell loaded with the appropriate weight plates. Participants were instructed to keep constant downward pressure on the bar throughout the jump and were encouraged to move the resistance as fast as possible to achieve maximal power output with each trial. The bar was not allowed to leave the shoulders of the subject, with the trial being repeated if this requirement was not met. A minimum of two trials at each load were completed, with additional trials performed if both peak power and jump height were not within 5% of the previous jump squat. Adequate rest was enforced between all trials (3 min).

All jump squats were performed while the subject was standing on the force plate with a LPT (PT5A-150; Celesco Transducer Products, Chatsworth, CA) attached to the bar. The LPT was attached 10 cm to the left of the center of the bar to avoid any interference caused by movement of the head during the jump. The LPT was mounted above the subject, and the retraction tension of the LPT (equivalent to 8 N) was accounted for in all calculations. Analog signals from the force plate and LPT were collected for every trial at 1000 Hz and analyzed using custom programs designed using LabVIEW software. The signal from the LPT was filtered using a fourth-order, low-pass Butterworth digital filter with a cutoff frequency of 10 Hz, and the voltage output was converted into displacement using laboratory calibrations. The vertical velocity of the movement was determined using a first-order derivative of the displacement data. Power output was calculated as the product of the vertical velocity and the vertical ground reaction force data. Acceleration of the movement was calculated using a second-order derivative of the displacement data and smoothed using a fourth-order, low-pass Butterworth digital filter with a cutoff frequency of 10 Hz. These data collection and analysis methodology have been validated previously (9), and the test-retest reliability for all jump variables examined was consistently $r \geq 0.90$.

A series of performance variables was assessed during the jump squats. Peak force, velocity, power, displacement,

and acceleration were determined as the respective maximal values achieved during the entire movement. Net impulse was assessed as the integral of vertical ground reaction force during the period of application in which force exceeded that required during stationary standing (i.e., above body weight). Rate of force development (RFD) was determined between the minimum and maximum force that occurred throughout the movement. Similarly, rate of power development (RPD) was also determined between the minimum and maximum power that occurred throughout the movement. Average power output was calculated during the concentric phase of the movement (i.e., time between minimum displacement and takeoff). Instantaneous force and velocity output at the time at which peak power occurred was also examined and was termed force at peak power and velocity at peak power, respectively. These values across each of the loads examined were used to generate the force–velocity and force–power relationships for the jump squat. Velocity at takeoff was defined as the velocity of movement at the time at which the force output was first zero (i.e., when the toes first left the force plate). Furthermore, time to takeoff was determined as the time between the initiation of the countermovement (i.e., start of the eccentric phase) and the point that force was zero (i.e., end of the concentric phase—takeoff).

In addition to these instantaneous performance variables, analyses of parameters throughout the jump movement were conducted. The power–time and force–time curves from each individual subject were selected from the beginning of the eccentric phase (i.e., initiation of countermovement assessed as the time at which a change in velocity first occurred) through to the end of the concentric phase (i.e., at takeoff when force and power reached zero). The velocity–time and displacement–time curves were selected from the beginning of the eccentric phase to peak displacement (i.e., zero velocity). Using a custom-designed LabVIEW program, the number of samples in each individual curve was then modified to equal 500 samples by changing the time delta (dt) between samples and resampling the signal ($dt = \text{number of samples in the original signal} / 500$). The sampling frequency of the normalized signals was calculated according to the following equation:

$$\text{normalized sampling frequency (Hz)} = \frac{1 \text{ second}}{\left[\frac{\text{no. samples in original signal}}{\text{no. samples in normalized signal}} \right] \times \left[\frac{\text{seconds}}{\text{per sample}} \right]}$$

Consequently, the sampling frequency of the modified signals was then equivalent to 815 ± 154 Hz for the power–time and force–time curves and 538 ± 68 Hz for the velocity–time and displacement–time curves. This resampling allowed for each individual's power, force, velocity, and displacement curves to be expressed during equal periods of time (i.e., the 500 samples represented the relative time—from 0% to 100%—taken to complete the jump). In other words, the various data sets were normalized to total movement

time so that data could be pooled. Each sample of the normalized power–, force–, velocity–, and displacement–time curves was then averaged across subjects within the stronger, weaker, or control groups, resulting in averaged curves with high resolution (sampling frequency of 538–815 Hz). This allowed for power, force, velocity, and displacement throughout the jump to be compared across baseline, midtest, and posttest as well as between groups. Intraclass test–retest reliabilities for power–, force–, velocity–, and displacement–time curves during the CMJ have consistently been $r \geq 0.94$, $r \geq 0.90$, $r \geq 0.89$, and $r \geq 0.92$, respectively, using this methodology (10).

Before assessment of sprint performance, subjects performed a warm-up consisting of 5 min of light jogging and three submaximal 20-m sprints. The sprint test was initiated from a standing start involving a staggered stance with the same front–back leg orientation used for every trial throughout the baseline, midtest, and posttest. Subjects were instructed to commence the sprint at will and accelerate as quickly as possible throughout the 40 m. Three trials were performed with each separated by a 3-min recovery. A series of six dual-beam timing gates (Speedlight; Swift Sports, Lismore, NSW, Australia) was used to record instantaneous time at 5, 10, 20, 30, and 40 m (timing was commenced when subject passed through a dual-beam timing gate positioned at their front foot). Flying 5 m was calculated as the time between the 5- and 10-m gates, whereas flying 15 m was calculated at the time between the 5- and 20-m gates. Intraclass test–retest reliability for all sprint performance variables examined was consistently $r \geq 0.90$.

EMG of the VL, vastus medialis (VM), and biceps femoris (BF) was collected on the dominant leg during the isometric squat and all jump squats. Disposable surface electrodes (self-adhesive Ag/AgCl snap electrode, 2-cm interelectrode distance, 1-cm circular conductive area; product 272; Noraxon USA, Inc., Scottsdale, AZ) were attached to the skin over the belly of each measured muscle, distal to the motor point, and parallel to the direction of muscle fibers. A reference electrode was placed on the patella. The exact location of the electrodes relative to the anatomical landmarks was marked on a sheet of tracing paper after the first testing session to ensure consistent placement in subsequent tests. Each site was shaved, gently abraded, and cleansed with alcohol before electrode placement to minimize skin impedance. Raw EMG signals were collected at 1000 Hz and amplified (gain = 1000, bandwidth frequency = 10–1000 Hz, input impedance < 5 k Ω ; Model 12D-16-OS Neurodata Amplifier System; Grass Technologies, West Warwick, RI). The amplified myoelectric signal was collected simultaneously with force plate and LPT data using a data acquisition system including an analog-to-digital card. Custom programs designed using LabVIEW software were used for recording and analyzing the data. The signal was full-wave-rectified and filtered using a dual-pass, sixth-order, 10- to 250-Hz band-pass Butterworth filter as well as a notch filter at 50 Hz. A linear envelope was created using

a low-pass, fourth-order Butterworth digital filter with a cutoff frequency of 6 Hz. Maximal voluntary contraction (MVC) for all muscles were determined by averaging the integrated EMG signal during a 1-s period of sustained maximal force output after the initial peak in the force curve during the isometric squat (intraclass test-retest reliability consistently $r \geq 0.91$). EMG activity during jumping was analyzed by averaging the integrated EMG signal from the beginning of the eccentric phase to takeoff. To standardize for the time taken to complete a jump squat, this value was then divided by the time to takeoff. The average integrated EMG (AvgIEMG) was then normalized by expressing it relative to the MVC. This is similar to methods previously used when comparing EMG between movements with different time components (29). The rate of rise in AvgIEMG (expressed as a percentage of MVC per second) was assessed during the 0% 1RM jump squat as the rate of change between minimum and maximum AvgIEMG throughout the movement (i.e., from the initiation of the countermovement to takeoff). Intraclass test-retest reliability for all EMG variables examined was consistently $r \geq 0.80$.

In vivo muscle architecture was assessed by B-mode ultrasonography recorded using an ultrasound console (SSD-1000; Aloka Incorporated, Tokyo, Japan) with a 7.5-MHz, 9-cm linear probe. The same experienced examiner completed all scans across the baseline, midtesting, and posttesting occasions. Scans were performed on the VL of the dominant leg, with subjects lying supine and their leg muscles completely relaxed. To assist with acoustic coupling, water-soluble transmission gel was applied to the transducer. Measurements were taken at 50% of the thigh length calculated as half the distance between the centers of the greater trochanter to the lateral condyle of the femur. Longitudinal images were obtained with the transducer oriented parallel to the muscle fascicles and perpendicular to the skin. The impact of discrepancies in transducer location and orientation on architectural differences observed between baseline, midtest, and posttest was minimized using several techniques. The location and two-dimensional orientation of the transducer relative to anatomical landmarks were mapped onto a sheet of tracing paper to ensure that the same site was used across all testing occasions. In addition, in an effort to minimize any differences in the three-dimensional orientation of the transducer between testing occasions, the live, onscreen image during midtest and posttest was compared with those taken during baseline testing. This allowed the examiner to match the unique features of the ultrasound images (i.e., heterogeneities in the subcutaneous adipose tissue and echoes from interspaces among the fascicles). These procedures have been used previously in literature, monitoring the impact of training on muscle architecture (3). Images were digitally recorded, and the superficial and deep aponeuroses were identified, with muscle thickness measured as the distance between the aponeuroses (3,20). Pennation angle was defined as the angle at which the fascicles arose from the deep aponeurosis (20). Intraclass test-retest reliabilities for muscle thickness and

pennation angle were $r = 0.95$ and $r = 0.90$, respectively. In addition, rater reliabilities for muscle thickness and pennation angle were $r = 0.98$ and $r = 0.93$, respectively.

Two-dimensional motion analysis was used to evaluate the movement mechanics during the squat 1RM and jump squats. A digital video camera (25 Hz; MV830i; Canon Australia Pty Ltd., North Ryde, NSW, Australia) was positioned 3.1 m from the subject, perpendicular to the subjects' sagittal plane. Dartfish software (Version 4.5 ProSuite; Dartfish, Sydney, NSW, Australia) was used to analyze the movements. The video footage was deinterlaced into fields, yielding a 50-Hz sampling frequency. During the jump squats, the minimum and maximum knee, hip, and ankle joint angles were assessed at the transition between the eccentric and concentric phases (minimum) and the last field before takeoff where the foot is still in contact with the force plate (maximum). The minimum knee, hip, and ankle angles were also assessed during the squat 1RM, representing the joint angles at the transition between the eccentric and concentric phases. Intraclass test-retest reliability and rater reliability for all joint angles assessed using motion analysis were consistently $r \geq 0.92$ and $r \geq 0.93$, respectively.

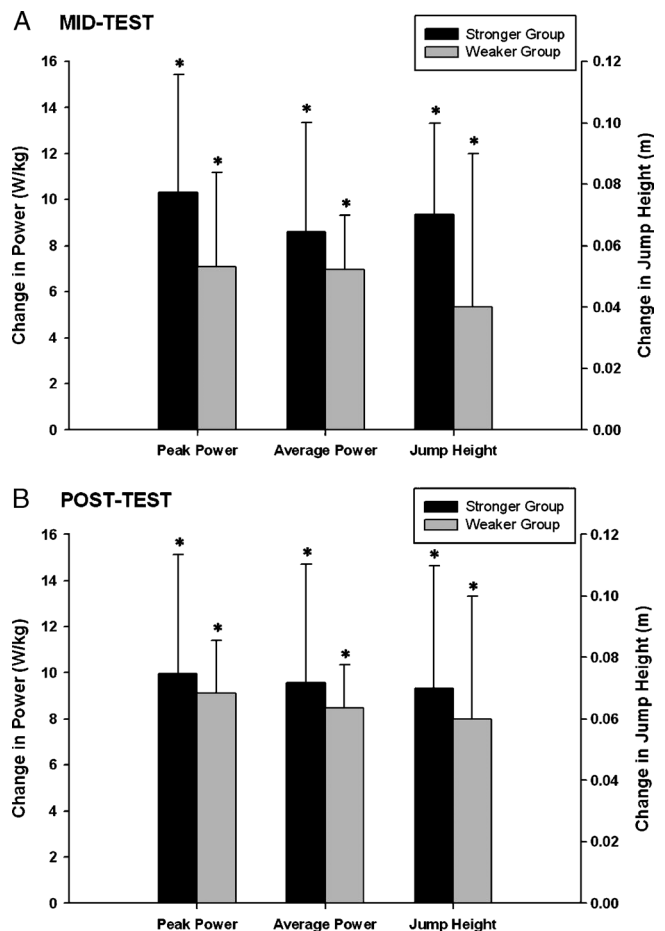


FIGURE 1—Change from baseline in peak power, average power, and maximum jump height during the 0% 1RM jump squat at midtest (A) and posttest (B). *Significant ($P \leq 0.05$) change from baseline.

TABLE 2. Change in performance variables during the 0% 1RM jump squat (i.e., BM only) from baseline (Δ).

0% 1RM Jump Squat	Stronger Group		Weaker Group	
	Δ at Midtest	Δ at Posttest	Δ at Midtest	Δ at Posttest
Power				
Peak power ($\text{W}\cdot\text{kg}^{-1}$)	$10.3 \pm 5.1^*$	$10.0 \pm 5.2^*$	$7.1 \pm 4.1^*$	$9.1 \pm 2.3^*$
Average power ($\text{W}\cdot\text{kg}^{-1}$)	$8.6 \pm 4.8^*$	$9.6 \pm 5.2^*$	$7.0 \pm 2.3^*$	$8.5 \pm 1.9^*$
Rate of power development ($\text{W}\cdot\text{kg}^{-1}\cdot\text{s}^{-1}$)	$343.0 \pm 164.4^*\ddagger$	$325.2 \pm 132.6^*$	$169.3 \pm 79.4^*$	$236.7 \pm 129.7^*$
Force				
Peak force ($\text{N}\cdot\text{kg}^{-1}$)	$6.3 \pm 5.4^*$	$6.3 \pm 4.7^*$	3.5 ± 3.7	$4.7 \pm 4.2^*$
RFD ($\text{N}\cdot\text{kg}^{-1}\cdot\text{s}^{-1}$)	$76.8 \pm 46.1^*$	$82.3 \pm 58.3^*$	$55.3 \pm 28.8^*$	$94.6 \pm 74.9^*$
Force at peak power ($\text{N}\cdot\text{kg}^{-1}$)	$2.8 \pm 2.1^*$	$2.1 \pm 1.6^*$	1.5 ± 1.8	1.4 ± 1.9
Net impulse (N·s)	$38.9 \pm 22.6^*$	$45.9 \pm 28.9^*$	$48.8 \pm 26.5^*$	$57.4 \pm 20.9^*$
Velocity				
Peak velocity ($\text{m}\cdot\text{s}^{-1}$)	$0.43 \pm 0.29^*$	$0.42 \pm 0.22^*$	$0.34 \pm 0.34^*$	$0.55 \pm 0.26^*$
Velocity at peak power ($\text{m}\cdot\text{s}^{-1}$)	0.19 ± 0.34	0.22 ± 0.31	0.15 ± 0.30	0.28 ± 0.27
Velocity at takeoff ($\text{m}\cdot\text{s}^{-1}$)	$0.43 \pm 0.29^*$	$0.41 \pm 0.24^*$	$0.35 \pm 0.36^*$	$0.60 \pm 0.30^*$
Acceleration				
Peak acceleration ($\text{m}\cdot\text{s}^{-1}\cdot\text{s}^{-1}$)	$11.7 \pm 6.2^*\ddagger$	$14.6 \pm 6.5^*\ddagger$	$6.4 \pm 2.0^*$	$8.6 \pm 4.7^*$
Displacement				
Peak displacement (m)	$0.07 \pm 0.03^*$	$0.07 \pm 0.04^*$	0.04 ± 0.05	$0.06 \pm 0.04^*$
Time				
Time to takeoff (s)	$-0.18 \pm 0.07^*$	$-0.19 \pm 0.6^*$	$-0.27 \pm 0.12^*$	$-0.31 \pm 0.13^*\ddagger$

* Significant ($P \leq 0.05$) change from baseline.

$\ddagger$ Significant ($P \leq 0.05$) difference between stronger group and weaker group.

Statistical analyses. A general linear model with repeated-measures ANOVA followed by Bonferroni *post hoc* tests was used to examine the impact of training on performance variables and to determine whether differences existed between the groups at baseline, at midtest, and at posttest. *t*-tests were used for comparisons between variables at baseline and posttest for the control group as well as for comparison of variables between stronger and weaker groups at midtest. Statistical significance for all analyses was defined by $P \leq 0.05$, and results were summarized as means $\pm$ SD. Estimated effect sizes (ES) of $\eta^2 = 0.502$ and $\eta^2 = 0.300$ at observed power levels of 0.976 and 0.626 for

maximum peak power relative to BM after training existed for stronger and weaker groups, respectively. In addition, estimated ES of $\eta^2 = 0.406$ and $\eta^2 = 0.319$ at the observed power levels of 0.821 and 0.644 existed for the comparison of maximum peak power relative to BM between the stronger and weaker groups at baseline and posttest, respectively. Mean ES were also calculated to examine and compare the practical significance of the performance improvements among the experimental groups. Based on the study of Cohen (8), which suggests ES of 0.2, 0.5, and 0.8 to represent small, moderate, and large effects, respectively, practical relevance was defined as an $\text{ES} \geq 0.8$ for the

TABLE 3. Comparison of 40-m sprint times among stronger, weaker, and control groups throughout the 10 wk of training.

	Sprint Times (s)						
	5 m	10 m	20 m	30 m	40 m	Flying 5 m	Flying 15 m
Stronger group							
Baseline	1.10 ± 0.10	1.82 ± 0.12	$3.08 \pm 0.15\ddagger$	$4.25 \pm 0.19\ddagger$	$5.41 \pm 0.25\ddagger$	$0.72 \pm 0.03\ddagger$	$1.98 \pm 0.07\ddagger$
Midtest	$1.08 \pm 0.02\ddagger$	$1.82 \pm 0.04\ddagger$	$3.07 \pm 0.09\ddagger$	$4.23 \pm 0.14\ddagger$	$5.39 \pm 0.20\ddagger$	0.74 ± 0.03	$1.99 \pm 0.08\ddagger$
Posttest	$1.02 \pm 0.08\ddagger$	$1.75 \pm 0.08\ddagger$	$2.99 \pm 0.11\ddagger$	$4.14 \pm 0.15\ddagger$	$5.29 \pm 0.20\ddagger$	$0.73 \pm 0.01\ddagger$	$1.97 \pm 0.05\ddagger$
Δ between baseline and posttest	$-0.08 \pm 0.04^*\ddagger$	$-0.07 \pm 0.04^*$	$-0.09 \pm 0.08^*\ddagger$	$-0.11 \pm 0.11^*\ddagger$	$-0.12 \pm 0.11^*$	0.01 ± 0.02	-0.01 ± 0.07
ES	0.86	0.67	0.69	0.65	0.54	0.64	0.18
Weaker group							
Baseline	1.17 ± 0.12	1.95 ± 0.16	3.32 ± 0.19	4.60 ± 0.23	5.90 ± 0.27	0.78 ± 0.04	2.14 ± 0.08
Midtest	1.13 ± 0.05	1.90 ± 0.08	3.24 ± 0.11	4.51 ± 0.16	5.79 ± 0.20	0.77 ± 0.05	2.10 ± 0.09
Posttest	1.09 ± 0.05	1.86 ± 0.07	3.18 ± 0.12	4.44 ± 0.17	5.69 ± 0.22	0.77 ± 0.02	2.09 ± 0.08
Δ between baseline and posttest	-0.08 ± 0.09	-0.09 ± 0.10	$-0.13 \pm 0.09^*\ddagger$	$-0.16 \pm 0.10^*\ddagger$	$-0.22 \pm 0.14^*\ddagger$	-0.01 ± 0.03	$-0.06 \pm 0.03^*\ddagger$
ES	0.79	0.69	0.78	0.76	0.82	0.25	0.66
Control group							
Baseline	1.13 ± 0.04	1.88 ± 0.07	3.25 ± 0.11	4.52 ± 0.14	5.79 ± 0.19	0.75 ± 0.03	2.12 ± 0.08
Posttest	1.15 ± 0.04	1.91 ± 0.05	3.27 ± 0.07	4.53 ± 0.09	5.79 ± 0.12	0.76 ± 0.02	2.12 ± 0.05
Δ between baseline and posttest	0.02 ± 0.05	0.03 ± 0.06	0.02 ± 0.08	0.01 ± 0.08	0.00 ± 0.10	0.01 ± 0.02	0.00 ± 0.04
ES	0.41	0.46	0.24	0.10	0.01	0.40	0.07

Flying 5 m time is calculated as the time between 5 and 10 m.

Flying 15 m time is calculated as the time between 5 and 20 m.

The change in sprint times between baseline and posttests as well as ES of the change is also displayed.

* Significant ($P \leq 0.05$) change from baseline to posttest.

$\ddagger$ Significant ($P \leq 0.05$) difference between stronger group and all other groups.

$\ddagger$ Significant ($P \leq 0.05$) difference between stronger group and weaker group.

$\ddagger$ Significantly ($P \leq 0.05$) different from control group.

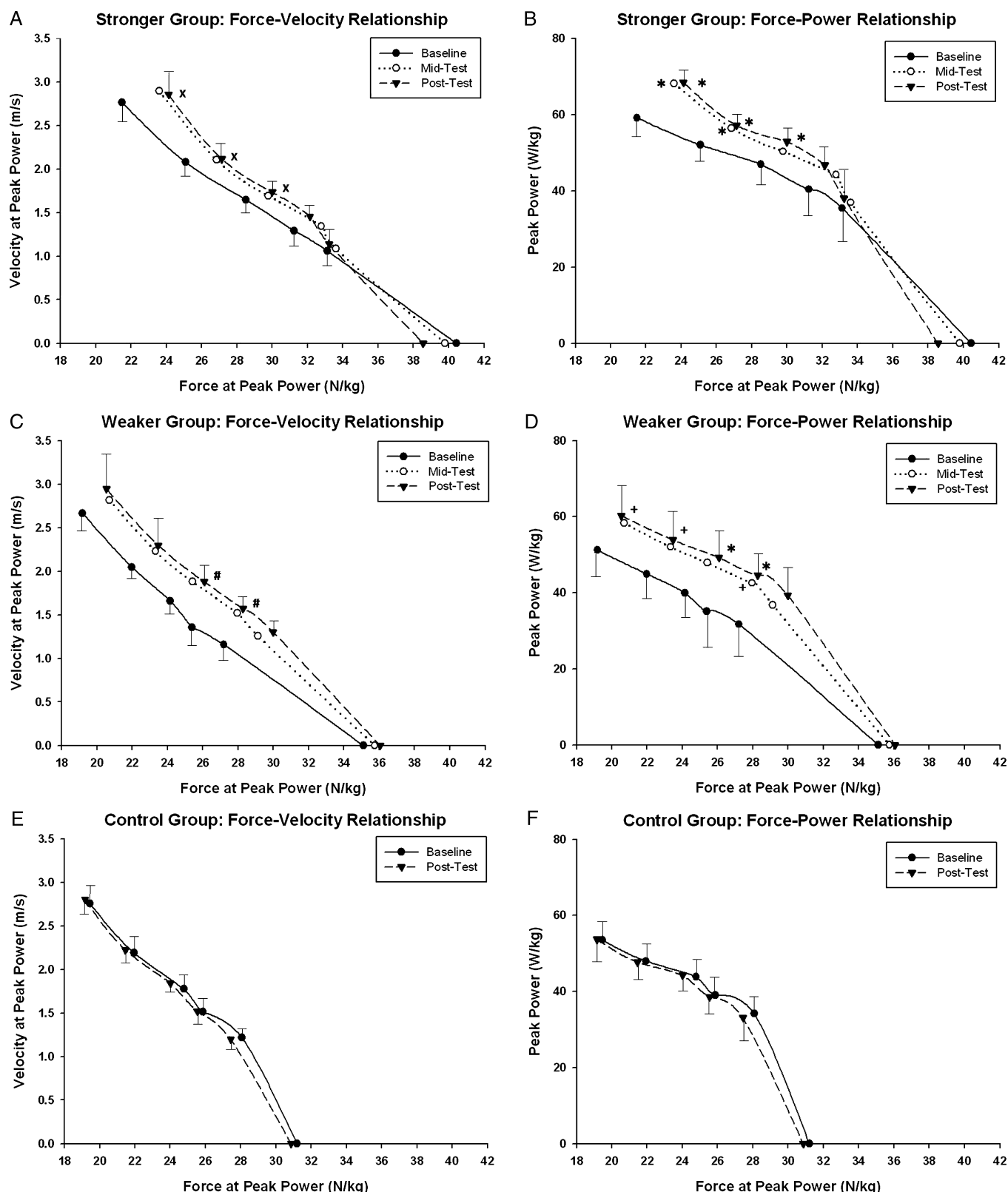


FIGURE 2—Training-induced changes to the force-velocity and force-power relationships for the jump squat. *Significantly ($P \leq 0.05$) different peak power relative to BM from baseline. +Approaching significantly ($P \leq 0.10$) different peak power relative to BM from baseline. xSignificantly ($P \leq 0.05$) different force relative to BM at peak power from baseline. #Significantly ($P \leq 0.05$) different velocity at peak power from baseline.

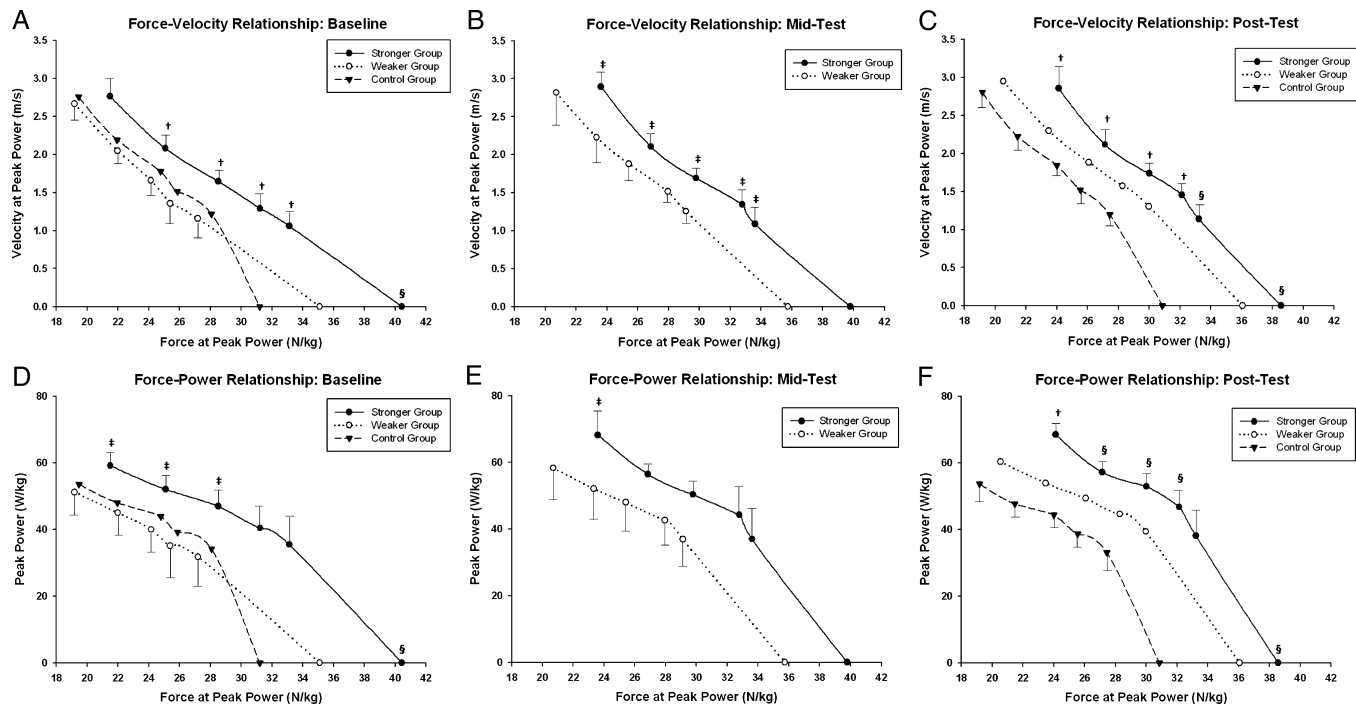


FIGURE 3—Between-group differences in the force–velocity (*A–C*) and force–power relationships (*D–F*) for the jump squat throughout the 10 wk of training. †Significant ($P \leq 0.05$) difference in force or power between stronger and all other groups. ‡Significant ($P \leq 0.05$) difference in force or power between stronger and weaker groups. §Significantly ($P \leq 0.05$) different force or power from control group.

purpose of this study. A statistical software package (SPSS, Version 13.0; SPSS, Inc., Chicago, IL) was used to perform all statistical analyses.

RESULTS

Athletic performance. The stronger group had significantly greater 1RM/BM than both the weaker and control groups at all testing occasions and displayed a practically relevant decrease in 1RM/BM at posttest ($ES = 0.93$, equivalent to a 7 ± 7 -kg decrease in 1RM). Training resulted in significant within-group changes in a multitude of jump performance variables between baseline and midtest or between baseline and posttest sessions for both the stronger and weaker groups (Fig. 1 and Table 2). However, only the changes in RPD at midtest and time to takeoff at posttest significantly differed between the groups (Table 2). Comparison of the magnitude of changes in average power, peak

power, and peak displacement between the stronger and weaker groups is illustrated in Figure 1. Any practically relevant differences between the groups were more pronounced after 5 wk of training, and they diminished somewhat after 10 wk of training (Fig. 1). The stronger group completed the 40-m sprint in significantly less time than both the weaker and control groups across all testing occasions (Table 3). Training resulted in a significant change in time between baseline and posttest at 5-, 10-, 20-, 30-, and 40-m sprints for the stronger group and at 20-, 30-, and 40-m sprints for the weaker group (Table 3). These changes were significantly different from the change in time displayed by the control group (which did not significantly alter sprint time).

Force–velocity relationship. Training-induced changes to force–velocity and force–power relationships during the jump squat are displayed in Figure 2. Significant differences between baseline and posttest were evident for

TABLE 4. Two-dimensional motion analysis variables assessed during the 0% 1RM jump squat.

0% 1RM Jump Squat	Minimum Joint Angle			Maximum Joint Angle		
	Knee Angle (°)	Hip Angle (°)	Ankle Angle (°)	Knee Angle (°)	Hip Angle (°)	Ankle Angle (°)
Stronger group						
Baseline	89.7 ± 8.5	86.3 ± 13.8	60.7 ± 4.4	170.0 ± 7.1	168.3 ± 4.9	133.4 ± 3.9
Midtest	94.8 ± 6.5	89.4 ± 10.8	61.8 ± 4.7	174.5 ± 5.0	171.9 ± 8.1	137.5 ± 5.9
Posttest	94.7 ± 5.6	91.5 ± 16.4	59.0 ± 5.4	176.8 ± 4.3	174.0 ± 7.1	140.0 ± 5.8
Weaker group						
Baseline	85.9 ± 7.1	79.3 ± 8.8	60.1 ± 3.0	170.2 ± 6.4	168.1 ± 8.1	136.0 ± 8.2
Midtest	90.1 ± 9.5	84.8 ± 18.7	59.9 ± 3.0	174.8 ± 4.9	172.8 ± 7.1	137.4 ± 6.6
Posttest	89.3 ± 11.1	88.4 ± 20.4	57.7 ± 3.7	171.7 ± 6.3	172.7 ± 9.5	138.1 ± 5.2
Control group						
Baseline	89.6 ± 6.4	78.6 ± 9.8	67.3 ± 6.3	174.3 ± 6.5	172.3 ± 4.9	141.4 ± 8.5
Posttest	86.0 ± 12.2	78.0 ± 6.9	66.3 ± 3.5	172.3 ± 9.3	170.9 ± 12.2	137.3 ± 7.0

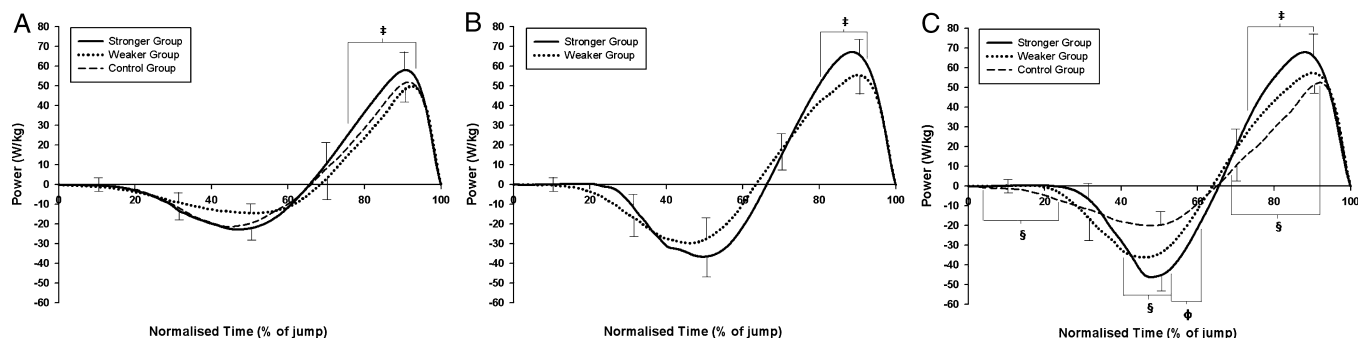


FIGURE 4—Comparison of the power–time curve during the 0% 1RM jump squat among the stronger, weaker, and control groups at baseline (A), midtest (B), and posttest (C). Normalized time represents the time from initiation of countermovement to takeoff in A, B, and C. †Significant ($P \leq 0.05$) difference between stronger and weaker groups. §Significant ($P \leq 0.05$) difference between control and both stronger and weaker groups. ΦSignificant ($P \leq 0.05$) difference between stronger and control groups.

both stronger and weaker groups (Fig. 2). Between-group comparisons of the force–velocity and force–power relationships during jump squats revealed several differences (Fig. 3).

Jump mechanics. No significant differences existed between the groups in either the minimum or the maximum joint angles during the 0% 1RM jump squat at baseline, midtest, or posttest (Table 4). Furthermore, training did not

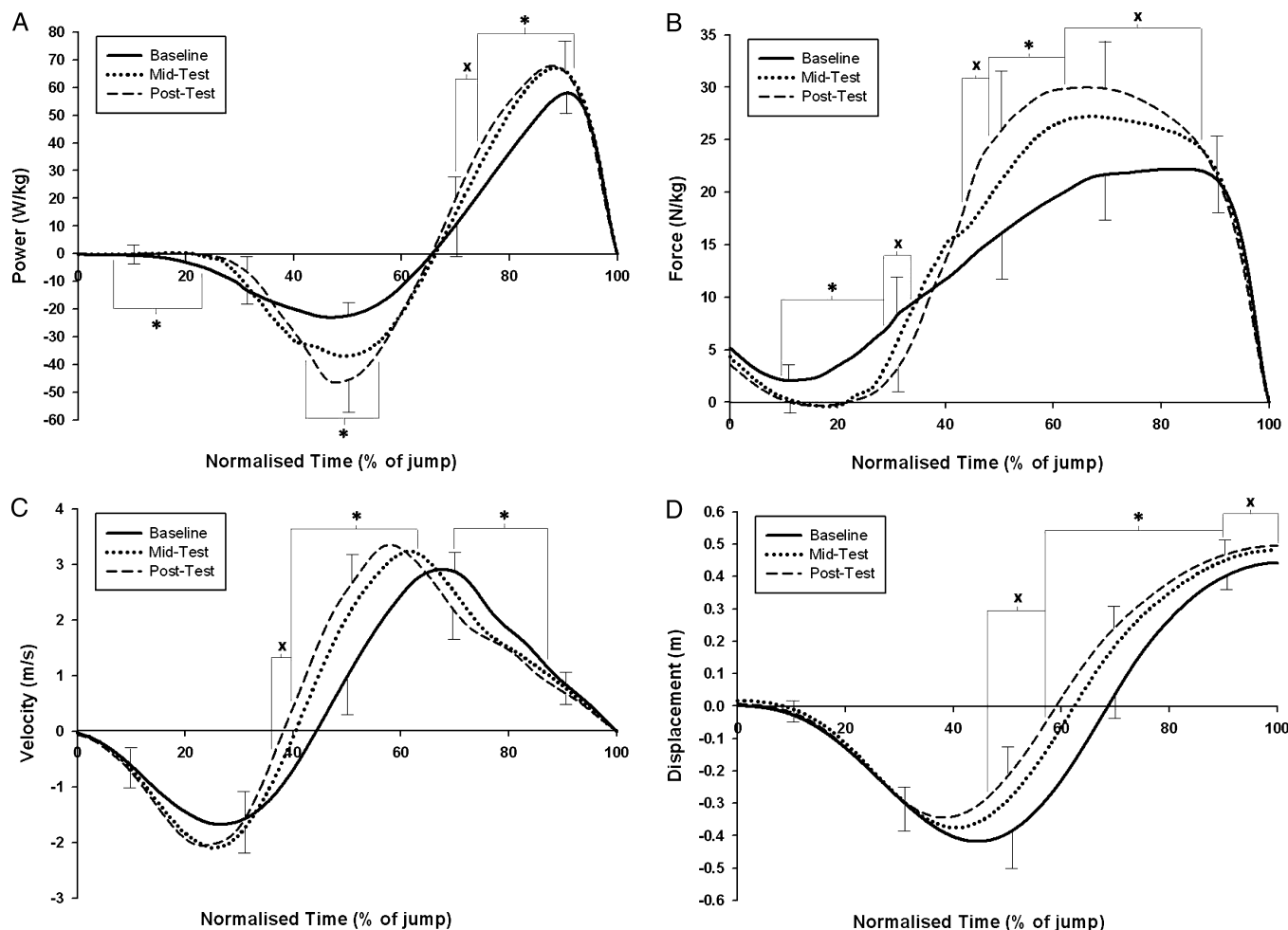


FIGURE 5—Training-induced changes to the power–time (A), force–time (B), velocity–time (C), and displacement–time (D) curves for the 0% 1RM jump squat in the stronger group. Normalized time represents the time from initiation of countermovement to takeoff for power and force (A and B) and from initiation of countermovement to peak displacement for velocity and displacement (C and D). *Significant ($P \leq 0.05$) difference between baseline and both midtest and posttest. xSignificant ($P \leq 0.05$) difference between baseline and posttest.

result in any changes to the joint angles at either the transition between eccentric and concentric phases or the take-off during the 0% 1RM jump squat (Table 4). Investigation of the power-time curve throughout the jump revealed significant differences between the groups (Fig. 4). At baseline, significant differences in power between stronger and weaker groups existed from 76.0% to 92.8% of normalized time (Fig. 4A). Similar differences between the stronger and weaker groups were maintained throughout midtest (80.4%–91.0% normalized time; Fig. 4B) as well as posttest (73.8%–90.2% normalized time; Fig. 4C). After training, significant differences were evident between the control group and both the stronger and weaker training groups throughout the power-time curve at posttest (4.4%–23.4%, 40.8%–52.8%, and 69.4%–89.2% normalized time). In addition, differences existed between the control and stronger groups from 53.0% to 60.8% normalized time (Fig. 4C). Several significant training-induced changes were also observed between baseline and both midtesting and posttesting sessions (Figs. 5 and 6). For the stronger group, significant differences between baseline and both midtest and posttest

existed during the following phases: (a) power: 6.6%–22.8%, 42.2%–55.6%, and 74.2%–91.2% of normalized time; (b) force: 9.6%–28.4% and 48.8%–62.8% of normalized time; (c) velocity: 39.6%–62.6% and 69.6%–86.8% of normalized time; and (d) displacement: 57.4%–89.4% of normalized time. Significant differences between baseline and posttest also existed during the following phases: (a) power: 70.6%–74.0% of normalized time; (b) force: 28.6%–32.4%, 43.0%–48.6%, and 63.0%–87.0% of normalized time; (c) velocity: 36.0%–39.4% of normalized time; and (d) displacement: 46.2%–57.2% and 89.6%–100.0% of normalized time (Fig. 5). For the weaker group, significant differences between baseline and both midtest and posttest existed during the following phases: (a) power: 3.6%–11.4%, 35.4%–52.6%, and 64.8%–89.6% of normalized time; (b) force: 0.0%–23.2% and 38.0%–70.0% of normalized time; (c) velocity: 9.4%–32.0%, 39.2%–68.0%, and 75.0%–91.8% of normalized time; and (d) displacement: 58.0%–88.2% or normalized time. Significant differences between baseline and posttest also existed during the following phases: (a) power: 11.6%–18.2% of normalized time, (b) force: 70.2%–79.2%

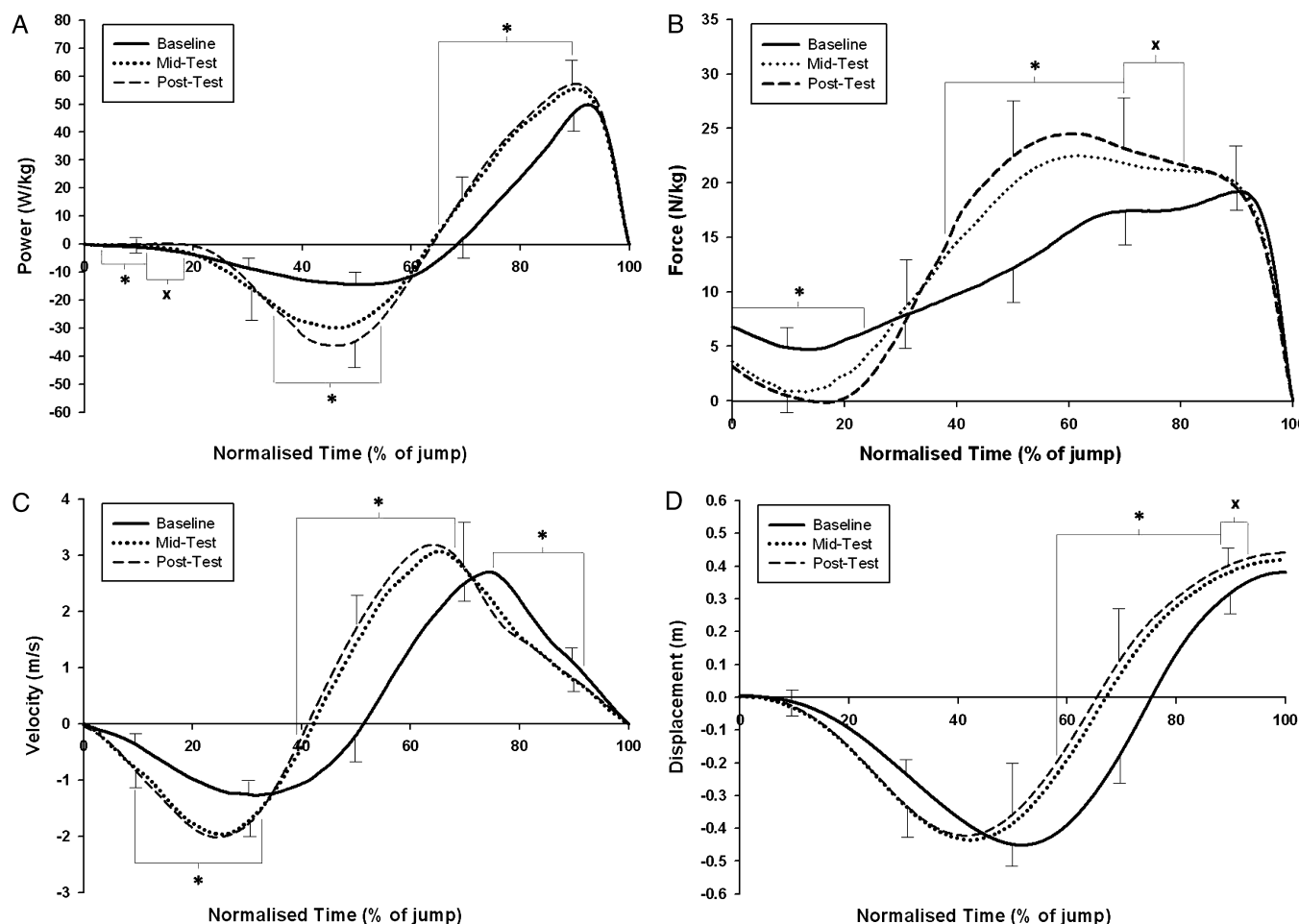


FIGURE 6—Training-induced changes to the power-time (A), force-time (B), velocity-time (C), and displacement-time (D) curves for the 0% 1RM jump squat in the weaker group. Normalized time represents the time from initiation of countermovement to takeoff for power and force (A and B) and from initiation of countermovement to peak displacement for velocity and displacement (C and D). *Significant ($P \leq 0.05$) difference between baseline and both midtest and posttest. xSignificant ($P \leq 0.05$) difference between baseline and posttest.

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TABLE 5. Neuromuscular factors assessed throughout the 10-wk training program.

	Stronger Group			Weaker Group			Control Group	
	Baseline	Midtest	Posttest	Baseline	Midtest	Posttest	Baseline	Posttest
Muscle architecture								
Muscle thickness (cm)	3.29 ± 0.38	3.30 ± 0.38	3.26 ± 0.37	3.02 ± 0.45	3.07 ± 0.46	3.08 ± 0.48	3.06 ± 0.30	3.06 ± 0.29
Pennation angle (°)	19.80 ± 2.23	20.76 ± 2.15	21.03 ± 2.14	19.20 ± 1.92	20.15 ± 1.59	20.98 ± 1.71	20.44 ± 3.99	20.59 ± 4.17
Lean mass of the leg (kg)	11.35 ± 1.62	11.56 ± 1.69	11.50 ± 1.62	10.82 ± 1.54	10.76 ± 1.50	10.66 ± 1.55	11.28 ± 1.61	11.30 ± 1.49
Muscle activation								
AvgIEMG IS (MVC)								
VM (μV)	81.9 ± 32.8	64.4 ± 24.6	68.0 ± 30.5	69.0 ± 28.9	82.3 ± 44.7	81.7 ± 28.2	79.6 ± 17.1	78.5 ± 18.2
VL (μV)	69.5 ± 22.7	66.6 ± 12.8	58.8 ± 26.9	73.3 ± 39.0	72.6 ± 42.8	71.2 ± 28.3	71.3 ± 11.9	70.0 ± 23.1
BF (μV)	13.9 ± 5.9	14.2 ± 11.9	13.8 ± 6.7	18.6 ± 14.2	14.3 ± 9.4	14.8 ± 10.2	20.3 ± 11.1	20.9 ± 12.0
AvgIEMG 0% 1RM JS								
VM (% MVC·s ⁻¹)	114.5 ± 24.2	145.2 ± 25.8	148.0 ± 45.3	108.8 ± 29.0	121.9 ± 52.3	115.3 ± 19.0	81.3 ± 13.6	84.6 ± 23.2
VL (% MVC·s ⁻¹)	115.2 ± 28.3	122.3 ± 20.0	135.0 ± 28.5	118.8 ± 46.8	130.1 ± 59.3	106.0 ± 25.7	107.1 ± 30.0	109.7 ± 28.5
BF (% MVC·s ⁻¹)	125.6 ± 21.8	195.5 ± 102.0	200.9 ± 112.5	100.8 ± 58.8	181.7 ± 73.8	174.4 ± 90.3	103.0 ± 53.2	104.9 ± 76.8
Rate of EMG rise 0% 1RM JS								
VM (% MVC·s ⁻¹)	503.5 ± 148.9	903.7 ± 380.9	1057.3 ± 492.0*	352.3 ± 174.4	725.1 ± 305.9*	672.5 ± 263.8*	373.0 ± 209.7	386.7 ± 227.4
VL (% MVC·s ⁻¹)	479.2 ± 210.2	634.1 ± 188.6	971.4 ± 338.6*	389.6 ± 187.4	866.0 ± 672.0	654.1 ± 283.0	403.6 ± 209.5	392.9 ± 146.8
BF (% MVC·s ⁻¹)	922.6 ± 413.9	1893.5 ± 1170.1	1759.3 ± 941.1	714.5 ± 551.9	1180.0 ± 619.0	1359.0 ± 984.6	449.7 ± 189.1	477.2 ± 250.8

AvgIEMG, average integrated EMG; IS, isometric squat; JS, jump squat.

Average integrated EMG (AvgIEMG) was assessed during an isometric squat MVC and during the 0% 1RM jump squat. Rate of rise in AvgIEMG was also assessed during the 0% 1RM jump squat. Muscle thickness and pennation angle were assessed through ultrasound images. Lean mass of the leg represents the average lean muscle mass of right and left legs as assessed through dual-energy x-ray absorptiometry scans.

* Significant ($P \leq 0.05$) change from baseline.

§ Significantly ($P \leq 0.05$) different from control group.

of normalized time, and (c) displacement: 88.4%–92.4% of normalized time (Fig. 6).

Neuromuscular characteristics. No between- or within-group differences existed for muscle thickness, pennation angle, or lean mass of the leg across each of the testing occasions (Table 5). However, percent change in pennation angle from baseline to posttest was significant (9.9%, $P = 0.05$) for the weaker group and approaching significance (6.5%, $P = 0.07$) for the stronger group. At baseline, no between-group differences were evident in any of the EMG variables assessed (Table 5). After training, no significant between- or within-group differences existed in AvgIEMG during the isometric squat (i.e., MVC) or the 0% 1RM jump squat. However, the stronger power training group significantly increased the rate of rise in AvgIEMG during the 0% 1RM jump squat at posttest in both VM and VL (Table 5). Furthermore, the rate of rise in AvgIEMG of the VM and VL in the stronger power training group was significantly greater than that in the control group at posttest. The weaker power training group also displayed a significant increase in the rate of rise in AvgIEMG of the VM during the 0% 1RM jump squat at both midtest and posttest (Table 5). No within-group changes in any EMG variables were observed for the control group throughout the study.

DISCUSSION

This investigation revealed that the ability to adapt to ballistic power training is quite similar for both strong and weak individuals. Despite trends toward superior improvements in maximal power production and athletic performance in stronger individuals (supported by much greater mean ES, especially after 5 wk of training), the magnitude of

improvements did not significantly differ between stronger and weaker groups (Fig. 1 and Tables 2 and 3). Furthermore, the mechanisms driving these adaptations were similar for both groups (Figs. 2–4 and Tables 4 and 5).

Athletic performance. The current results indicate that the experimental training effectively improved athletic performance in both stronger and weaker individuals. Jump height and maximal power output during a CMJ were significantly enhanced after both 5 and 10 wk of training (Fig. 1 and Table 2). In addition, a range of jump performance measures (i.e., net impulse, movement velocity, RFD, and RPD) also showed significant improvement in both experimental groups. These observations are similar to previous research involving ballistic power training on homogeneous groups of subjects with relatively low (11,38) or moderate strength levels (25,28). Comparisons of the magnitude of these improvements in jump performance between the experimental groups showed no statistically significant differences (Fig. 1 and Table 2). However, ES analyses revealed that practically relevant differences existed between stronger and weaker individuals in the magnitude of improvements in jump performance after ballistic power training. Any practical differences between the groups were more pronounced after 5 wk of training and diminished somewhat after 10 wk of training. For example, after 5 wk of training, the ES of improvements in CMJ peak power and jump height were 1.60 and 1.59, respectively, for the stronger group compared with 0.95 and 0.61 for the weaker group. At the completion of 10 wk of training, ES were 1.55 and 1.46 for the stronger group and 1.03 and 0.95 for the weaker group. Thus, despite the lack of statistically significant differences between the experimental groups, ballistic power training had a tendency to elicit a more pronounced effect on the magnitude of improvements of the stronger group, and

this could hold great practical relevance. Furthermore, these results suggest that stronger subjects also display a tendency toward more rapid improvements in performance after ballistic power training than weaker individuals, which is also of considerable practical importance.

Ballistic power training also resulted in enhanced sprint performance for both stronger and weaker groups (Table 3). The change in time between baseline and posttest was significant for both groups at 20, 30, and 40 m during a 40-m sprint. In addition, the stronger group also displayed a significant reduction in time at the 5- and 10-m marks. The improvements in sprint performance for both groups were of a practically relevant magnitude, representing a 7.3% and 2.2% improvement in 5- and 40-m time ($ES = 0.86$ and 0.54) in the stronger group and a 6.8% and 3.7% improvement in 5- and 40-m time ($ES = 0.79$ and 0.82) in the weaker group. Although the additional significant improvements of the stronger group (i.e., at 5 and 10 m) may indicate a greater adaptability to the training stimulus, there were no significant differences in the magnitude of performance enhancements between the stronger and weaker groups. Furthermore, the mean ES of the improvements in sprint performance were quite similar between the experimental groups (Table 3). Previous research has reported improvements in sprint performance approaching significance after ballistic power training (38), but such trends have not been consistently observed (25,37). Thus, the current findings are of great importance because this study provides the strongest evidence to date that sprint performance can be improved by ballistic power training in the form of vertical jumping. The successful transfer of jump squat training to sprint performance in the current study is theorized to be associated with the efficacy of the program design (i.e., highly sports-specific movements, frequency of training, effective load, repetition, set, and interset recovery parameters).

The ability of the current study to elucidate whether the magnitude of performance improvements after ballistic power training is influenced by strength level was limited by several factors. First, the principle of diminished returns dictates that initial improvements in muscular function are easily invoked and further improvements are progressively harder to achieve (37). Thus, the training programs of individuals with significantly greater strength levels (and more experienced training backgrounds) typically need to contain added variability, compared with weaker, inexperienced individuals (27,37). However, the nature of the research questions addressed by this study meant that the stronger and weaker groups completed the same training program during a period of 10 wk. Although the training program was designed to contain a great deal of specificity to common athletic movements, the experimental training lacked the level of variability required to maximize improvements in experienced athletes. Consequently, the ability of this intervention to maximize the adaptations of the stronger group may have been affected. Second, the current observations may have been

confounded by the stronger groups' cessation of strength training for the duration of the study (i.e., total of 15 wk). To examine the specific mechanisms driving adaptation to the ballistic power training intervention, all subjects were instructed to refrain from any lower body resistance training (or sprint training) outside the scope of the current study. Although this posed no impact on the strength level of the weaker group (i.e., not involved in any such training before commencing the experiment), the stronger group displayed a statistically nonsignificant but practically relevant decrease in maximal strength (4.6% decrease in 1RM/BM; $ES = 0.91$). The neuromuscular changes associated with detraining periods of a similar time course (i.e., decreased neural drive and CSA [15]) are theorized to negatively impact the ability of the stronger group to adapt to the experimental training. Thus, the cessation of strength training is theorized to have negatively affected the ability of the stronger group to adapt to the ballistic power training. It is imperative, therefore, that future research is conducted that incorporates strength maintenance sessions during the training intervention to elucidate further the influence of strength on the magnitude of improvements in athletic performance. Cognizant of these limitations, the fact that the stronger group showed similar performance improvements as the weaker group to the ballistic power training intervention is of great practical importance.

Mechanisms responsible for improved performance.

Monitoring the training-induced changes to the force-velocity and force-power relationships during a sports-specific movement such as the jump squat offers some indication of the mechanisms driving adaptations in the stronger and weaker groups. The ability of such an applied *in vivo* measure of the force-velocity relationship to delineate exact changes to muscle mechanics after training is complicated by a range of factors including mixed fiber composition, muscle architectural characteristics, anatomical joint configuration, levels of neural activation, as well as the complex nature of the jump squat movement (23). Despite these limitations, examination of the force-velocity relationship in sports-specific movements quantifies the ability of the intact neuromuscular system to function under various loading conditions, information essential to understanding muscular function during dynamic athletic movements. Data from this investigation revealed that strength level did somewhat influence the training-induced changes to the jump squat force-velocity and force-power relationships (Figs. 2 and 3). Specifically, the stronger group showed a more velocity-specific response to the training stimulus, displaying the greatest improvements under the lightest loading conditions (i.e., high-velocity, low-force portion of the force-velocity relationship). In contrast, ballistic power training with 0%–30% 1RM resulted in improvements in velocity and power throughout a range of loading conditions for the weaker subjects (i.e., similar improvements at both high-velocity and high-force portions of the force-velocity relationship). These changes caused the slope of the force-velocity relationship to increase in the

stronger group (aided by a slight decrease in $F_{\max}$) but not in the weaker group. Thus, the current data support the theory of velocity specificity (13,19) and the notion that relatively weak or inexperienced subjects display relatively nonspecific adaptations to training when compared with stronger, more experienced athletes. Although the stronger group seemed to display a more velocity-specific response, these apparent dissimilarities did not translate into major changes in the significant between-group differences that were evident before training (Fig. 3). However, a significant between-group difference in force at peak power during the lightest load examined (i.e., 0% 1RM) that did not exist at baseline was observed at midtest and at posttest. In addition, between-group differences in power at 20% and 40% 1RM loads that existed before training were no longer present at posttest. These training-induced changes to the significant differences between stronger and weaker groups also indicate that stronger individuals displayed adaptations with greater velocity specificity.

Another potential mechanism driving the observed improvements in athletic performance was changes to movement mechanics during jumping. Although no significant changes were observed in the minimum and maximum hip, knee, and ankle angles during the 0% 1RM jump squat (Table 4), significant training-induced changes in displacement, velocity, force, and power were evident throughout the 0% 1RM jump squat for both groups (Figs. 4–6). These changes are theorized to have led to an optimization of stretch-shorten cycle (SSC) function, which contributed to the enhanced jump performance. Arguably, the primary mechanism driving the enhancement of performance during SSC movements is the attainment of a greater level of force at the beginning of the concentric phase in comparison to concentric-only movements (5). Both stronger and weaker groups displayed significant increases in force during the eccentric phase and early in the concentric phase of the 0% 1RM jump squat after training (Figs. 5 and 6). This increase was generated by the improved acceleration of the BM during the eccentric phase and, similar to comparisons between SSC and concentric-only movements, resulted in greater net impulse, velocity of movement, power output, and, ultimately, enhanced jump height after training. The observed changes are hypothesized to be due to the slight modifications in jumping mechanics (i.e., a marginally shorter but faster countermovement) and a significant decrease in time to takeoff (Tables 3 and 4). Very little previous research exists examining the impact of training on performance variables throughout the entire movement however, similar results have been observed after an analogous ballistic power training intervention involving relatively untrained men (10). Comparisons between the stronger and the weaker groups revealed no new significant between-group differences after training in joint angles of the hip, knee, and ankle (Table 4) or power output throughout the 0% 1RM jump squat (Fig. 4). Thus, changes to jump mechanics common to both stronger and weaker

individuals after ballistic power training involving jump squats are theorized to have contributed to improvements in jump performance.

This investigation revealed that stronger and weaker subjects displayed similar adaptations in muscle architecture after ballistic power training (Table 5). As expected, the training intervention did not elicit any significant changes to muscle thickness (indicative of whole-muscle CSA [21]) or the lean mass of the leg for either experimental group. The relatively light loads used during ballistic power training (i.e., 0%–30% 1RM) were too small to elicit the necessary mechanical stimulus required to initiate a significant hypertrophic response (16,17). In addition, ballistic power training did not elicit any significant differences in the pennation angle of either the stronger or the weaker group. Changes in pennation angle have been reported previously, with increases observed after heavy resistance training (4,21), although not consistently (3), and decreases in response to sprint training (4). These changes are believed to have a positive impact on the force- and velocity-generating capacity of muscle, respectively. However, the potential impact of ballistic power training on changes to pennation angle has not been examined previously. On the basis of the current data, ballistic power training did not prompt structural changes to the muscle (i.e., muscle thickness or pennation angle). Furthermore, the initial strength level of the subject did not impact the type of adaptations in muscle architecture. Muscular adaptations at an intracellular level (i.e., alterations to anaerobic and aerobic enzymes, muscle substrates, and/or protein expression) and/or connective tissue remodeling may have contributed to the observed performance improvements (14). However, the potential for such adaptations cannot be established or rejected because these mechanisms were not assessed in the current study.

Neural adaptations in response to ballistic power training were observed, with significant changes evident in the neural activation patterns of subjects regardless of their initial strength level. Changes in EMG (indicative of alterations in motor unit recruitment, firing frequency, and/or synchronization) have been previously reported with improvements in performance after ballistic power training (16,25,36). However, this is one of the first experiments to show training-induced changes in EMG during complex, multijoint, sports-specific movements. The current data indicate that ballistic power training resulted in significant increases in the rate of EMG rise during dynamic athletic performance (i.e., 0% 1RM jump squat) in both stronger and weaker groups. Thus, it is theorized that the ballistic power training enhanced intermuscular coordination by optimizing the magnitude and timing of muscle activation. Similarly, Häkkinen et al. (16) observed ballistic power training (jump squats with 0%–60% 1RM) to result in a 38% increase in the rate of EMG rise during an isometric knee extension, which was reported to contribute to improved performance (a 24% improvement in isometric RFD). Although the current study cannot delineate whether

the observed changes in EMG were brought about through alterations in motor unit recruitment, firing frequency, and/or synchronization, previous research involving intramuscular EMG may offer some insight. During ballistic contractions, motor units have been reported to begin firing at very high frequencies (even in excess of those required to achieve maximal force) followed by a rapid decline (40). The high initial firing frequency is believed to result in increased RFD, even if only maintained for a very short period (26). van Cutsem et al. (36) reported the peak firing frequency at the onset of ballistic contraction to increase after ballistic power training. Furthermore, these higher firing frequencies were maintained for longer throughout the contraction after training. In addition, a training-induced increase in the percentage of doublet discharges (i.e., a motor unit firing two consecutive discharges in a ≤ 5 -ms interval) at the onset of a ballistic contraction was also reported (5.2% of motor units displayed doublet discharges before training, and this increased to 37.2% after 12 wk of ballistic power training). These training-induced changes were reported to contribute to an 82.3% increase in the RFD and a 15.9% improvement in time to peak force during ballistic contractions (36). Therefore, the ballistic power training intervention of the current study may have induced adaptations to the pattern of motor unit firing frequency that subsequently enhanced RFD capabilities and contributed to enhanced athletic performance. It is important to note that these mechanisms of adaptation as well as the performance improvements observed are specific to the movement pattern and loading parameters used in the current study as power produced by muscle varies according to both the nature of the movement and the loading parameters used (12).

In conclusion, the ballistic power training program used was very effective at enhancing athletic performance, with both groups showing significant improvements in maximal power, jump height, movement velocity, and sprint performance. The magnitude of improvements in athletic performance after ballistic power training did not significantly differ between strong and weak subjects. However, ES analyses revealed that the training had a tendency toward producing a more pronounced effect on jump performance in the stronger group (especially after only 5 wk of training) that is believed to hold great practical relevance. Further-

more, this conclusion is strengthened by the confounding influences of the principle of diminished returns as well as the stronger groups' cessation of strength training. Therefore, because stronger individuals display superior performance before ballistic power training and have a tendency for greater improvements after such training, it would be advantageous for individuals to establish a solid foundation of strength before focusing on ballistic power training. The mechanisms driving the performance improvements were very similar for both the stronger and the weaker groups. Ballistic power training involving sports-specific movements increased the rate of EMG rise during jumping, which, coupled with slight technique modifications believed to be specific to this training stimulus, led to an improvement in SSC function. As a result, subjects were able to achieve greater force and more optimally timed force application resulting in higher acceleration and movement velocity in shorter periods. Thus, athletic performance was improved through enhanced magnitude and RFD, translating to higher velocity and power production capabilities. Not only do these findings provide a deeper understanding of the mechanistic factors driving improvements in performance after ballistic power training but they also reveal that the neuromuscular and biomechanical adaptations to such training are not influenced by strength level. These findings have several implications for the design of ballistic power training programs that effectively improve athletic performance. The use of ballistic jump squats with very light loads (i.e., 0%–30% 1RM) is sufficient to induce significant improvements in jump and sprint performance of both strong and weak athletes. However, ballistic power training programs of stronger athletes need to contain considerably more variability than programs of weaker athletes for continued performance improvements beyond 5 wk of training. Finally, the incorporation of strength maintenance sessions throughout a power training phase is vital because decrements in maximal strength (and the ensuing neuromuscular alterations [15]) are theorized to negatively affect the ability of stronger athletes to adapt to ballistic power training.

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